

END TERM EXAMINATION

THIRD SEMESTER [BCA] JANUARY 2024

Paper Code: BCA-221

Subject: Principles of Management &
Organizational Behaviour

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit.

- Q1 Write short notes of the following (Any Five). (5x5=25)
- (a) Bureaucratic management approach
 - (b) Decision making
 - (c) Authority and responsibility
 - (d) Learning
 - (e) Perception
 - (f) Extrinsic and Intrinsic motivation

UNIT-I

- Q2 What do you mean by the term Management? Explain the role of a manager in an organisation. (12.5)
- Q3 Briefly explain the history of development of management thought.

UNIT-II

- Q4 Discuss the process of planning in details with examples. (12.5)
- Q5 What do you understand by Staffing? Briefly explain the components of staffing.

UNIT-III

- Q6 Discuss Maslow's Need Hierarchy Theory. How does the theory holds good in Indian context? (12.5)
- Q7 Explain the meaning and nature of organisational behaviour? What are the various challenges of organisational behaviour faced by managers in today's scenario?

UNIT-IV

- Q8 Briefly explain the Models of organisational behaviour. (12.5)
- Q9 Define Personality. Explain various determinants of Personality.

END TERM EXAMINATION

THIRD SEMESTER [BCA] JANUARY 2024

Paper Code: BCA-207

Subject: Human Values and Ethics

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

- Q1 Write short notes of the following (Any Five):- (5x5=25)
- (a) What do you mean by Holistic Approach? Explain it with an example.
 - (b) Differentiate between Recycling and Sustainability. Explain the term 'self regulation' in nature
 - (c) Explain the role of Professional Integrity and why it is important?
 - (d) How do you describe professional competence?
 - (e) Write a short note on Self Exploration
 - (f) What are choice making- Choosing, Cherishing and Acting in relation to right understanding.

UNIT-I

- Q2 (a) What are thought provoking issues in life? Explain the terms of happiness and prosperity. (8)
(b) Write a note on Self-Satisfaction. Explain the different tools for self exploration. (4.5)
- Q3 (a) Explain the concept of Value education? Define the basic need and process of Value education. (8)
(b) Differentiate between Spiritual values and Moral values. (4.5)

UNIT-II

- Q4 (a) Explain the four orders in nature. How are they interconnected and mutually fulfilling for each other. (8)
(b) Explain the importance of Coexistence in nature. (4.5)
- Q5 (a) Explain the concept of Vedanta in management. Briefly examine the features of Vedanta Ethics. (8)
(b) How is the Vedanta philosophy relevant at workplace? (4.5)

UNIT-III

- Q6 (a) Why a culture of openness in the workplace is essential? Write down the features of employee loyalty towards goals and objectives. (8)
(b) What are the best actionable practices for avoiding the fraudulent use of institutional resources? (4.5)
- Q7 (a) Write the functions of harmony in the society by universal human order and through undivided society. (8)
(b) How can we achieve harmony in our family community and in nation? (4.5)

UNIT-IV

- Q8 (a) Explain the term of Sexual misconduct and what kinds of misconduct can happen at work? (8)
(b) How can we promote equality and inclusion in the workplace? (4.5)
- Q9 (a) Explain the role of trust in the workplace and what causes lack of trust in an organization. (8)
(b) Write the importance of balancing privacy and transparency in the workplace. (4.5)

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END TERM EXAMINATION

THIRD SEMESTER [BCA] JANUARY 2024

Paper Code: BCA-201

Time: 3 Hours

Subject: Computer Networks

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory.
Select one question from each unit.

- Q1 Attempt any five of the following:- (5x5=25)
- a) Define a network. Discuss various Network Criteria.
 - b) Compare TCP and UDP protocols
 - c) Define Piggybacking and specify its benefit.
 - d) Write a short note on Modems.
 - e) Differentiate between Bridges and Routers.
 - f) In reference to UDP, what is the role of Pseudoheader and the Length field?
 - g) What is bit stuffing and byte stuffing?
 - h) Compare and contrast the different Unguided transmission media.

UNIT-I

- Q2 a) What is layered architecture? Explain the OSI model in brief. (6.5)
b) Explain LAN, MAN & WAN. (6)

OR

- Q3 a) Discuss the topologies in detail with examples. (7.5)
b) Explain the Shannon capacity theorem. (5)

UNIT- II

- Q4 a) What is understood by Flow Control? Explain any one Sliding Window ARQ protocol with diagrams and example. (6.5)
b) Given a 10-bit sequence 1010011110 and a divisor of 1011, find the CRC and check your answer. (6)

OR

- Q5 a) Discuss the sliding window protocol. Specify its advantages over stop and wait protocol. (6.5)
b) Distinguish the following (i) Synchronous and Asynchronous TDM (6)
(ii) Circuit switching and Packet switching

UNIT-III

- Q6 Explain the process of Distance vector routing with an example. Explain when the sharing happens and how routing tables are exchanged between routers. (12.5)

OR

- Q7 a) A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. Calculate: Frame size, Frame rate, Frame duration, Bit rate of link. (5)
b) What is understood by Classful addressing? Explain. What is its disadvantage? (7.5)

P.T.O.

UNIT- IV

- Q8 a) Explain the functions and protocols of the Transport Layer. (7.5)
b) Explain Connection Management in reference to transport layer. (5)
- OR
- Q9 Explain the Functions of Session layer, Presentation layer and Application layer. (12.5)

BCA-201
P_{1/2}

BCA-201
P_{2/2}

END TERM EXAMINATION

THIRD SEMESTER BCA JANUARY 2024

Paper Code: BCA-205

Subject: Object Oriented Programming
with C++

Time: 3 Hours

Note: Attempt five questions in all including Q1 which is compulsory. Select one question from each unit.

1 Answer the following (any five):

(5X5=25)

- What are different types of inheritance? Explain with the help of diagram and example.
- Distinguish between data abstraction, data encapsulation and data hiding with example.
- What is the difference between structures and classes in C++?
- What is the difference between constructor and destructor? Explain with example.
- How is polymorphism achieved at a) compile time and b) run time?
- What are the functions to open, close, read and write into a file. Give syntax.

UNIT - I

Write a menu driven program with the following options:

- Check whether a number is even / odd.
 - Check whether a number is divisible by another number.
 - Check whether a character entered is vowel or not.
- (Hint: use function overloading)

(12.5)

OR

(2.5x5=12.5)

Explain the following with example:

- Inline functions
- Reference variable
- New and delete operator
- Procedure and object oriented programming
- Object and instance

UNIT - II

Q4 What is a friend function? What are its various characteristics? Explain with example. Can we have a friend class? (12.5)

OR

Q5 Can we overload constructors? If yes, please explain different types of constructors with example. (12.5)

P.T.O.

UNIT - III

Q6 i) Write a menu driven program to overload arithmetic binary operators (+, -, *, /) using user defined type. (10)

ii) Explain different access specifiers with respect to inheritance. (2.5)

OR

Q7 a) WAP for conversion between built in and user defined data type. (7)
b) How to define virtual and pure virtual functions? Explain with a small segment of code. (5.5)

UNIT - IV

Q8 a) What is an exception? What is the procedure of handling exceptions in C++? How to specify an exception for a function? (8)
b) How generic programming is achieved in C++? How to define generic function? (4.5)

OR

Q9 Write an interactive program to create an employee file. It has to maintain information such as empid, name, qualification, designation and salary. The user must be able to access all details about a person either by entering employee name or by employee id. (12.5)

BCA-205
P2/2

BCA-205
P1/2

END TERM EXAMINATION

THIRD SEMESTER [BCA] JANUARY 2024

Paper Code: BCA-203

Subject: Computer Organization and Architecture

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

- Q1 Attempt the following (any five) [5x5=25]
- a) What are logic gates? Draw the schematic diagram of XOR gate. Give its truth table.
 - b) Draw a MUX using NAND gates which selects from four inputs A0 to A3 and two select inputs S0 and S1.
 - c) What is the role of stack pointer in computer organization?
 - d) What is ROM? How PROM, EPROM and EEPROM differ from each other?
 - e) Design and explain Ring counter.
 - f) Explain the edge triggered D flip-flop.

UNIT-I

- Q2 a) What are Universal gates? Explain how basic gates can be realized using NAND and NOR gates. [6.5]
b) State and prove Associative and Distributive theorems. [6]
- Q3 a) Draw a full subtractor circuit using NAND gate. [6.5]
b) Minimise the following Boolean function using K-map. [6]
 $F(A, B, C, D) = \Sigma (3, 4, 5, 7, 9, 13, 14, 15)$

UNIT-II

- Q4 a) Realize JK flip-flop using SR flip flop. [6.5]
b) Differentiate between flip-flop and latches. [6]
- Q5 a) Describe the operation of PISO shift register with the help of block diagram. [6.5]
b) Differentiate Combinational and Sequential circuits? [6]

UNIT-III

- Q6 a) What is instruction cycle? Draw detailed flowchart of the instruction cycle. [6.5]
b) What are CPU buses and Why they are important? [6]
- Q7 a) Explain the different types of addressing modes in basic computer. [6.5]
b) What is a register in a CPU and How Does it Work? [6]

UNIT-IV

- Q8 a) Write difference between Programmed Input /Output and Interrupt Driven Input/output. [6.5]
b) How DMA controller communicates and transfers data between peripheral devices and RAM. [6]

P.T.O.

- Q9 a) Draw and explain the memory hierarchy structure and mark the arrow from low to high (speed) and high to low (Cost). [6.5]
b) What is associative memory? Explain its working with the help of diagram. [6]

BCA-203
P1

BCA-203
P2/2

END TERM EXAMINATION

THIRD SEMESTER (BCA) JANUARY-2024

Paper Code: BCAT-211

Subject: Basic of Python Programming

Time: 3 Hours

Maximum Marks :75

Note: Attempt five questions in all including Q.no.1 which is compulsory.
Select one question from each unit.

- Q 1 Write short notes on the following: (Any Five) (5x5=25)
- (a) Elaborate - "Python is interpreted as well as integrated programming language"?
 - (b) How unary operator different from binary operator? Give example of each.
 - (c) Explain the use of `_init_` method in python programming?
 - (d) Differentiate between list and tuple?
 - (e) Explain the use of pandas library with an example? *
 - (f) Justify the significance of self keyword? Give an example?

UNIT-I

- Q2 (a) Explain while loop and for loop with an example. Elaborate difference between them. (4.5)
- (b) Explain membership operator, identity operator, escape sequence, slicing with an example? (8)
- Q3 (a) Perform following slicing operation on string "PYTHON PROGRAMMING":
`string[2:15:2]`,
`string[-3:-16:-2]`, `string[:13]`, `string[6:]`, `string[:-2]`, `string[3:14:3]` (6)
- (b) Write a program to print stars in right triangle pattern (value enter by user)? (6.5)

UNIT-II

- Q4 (a) Differentiates between mutable and immutable objects with examples? (4.5)
- (b) Explain tuple, list, sets, dictionary in python? Each with example. (8)
- Q5 (a) Explain list methods- append, extend, remove, index, pop, insert, sort? Give example of each? (6.5)
- (b) Explain dictionary methods- len, str, clear, copy, update, get? Give an example of each. (6)

UNIT-III

- Q6 Explain the concept of class, objects, functions, data member with help of program? Also Explain call by Value and call by reference? (12.5)
- Q7 Explain the following terms with examples: (12.5)
- (i) Scope and lifetime of a function.
 - (ii) Importing modules
 - (iii) Formal, Actual, Default, keyword arguments
 - (iv) Concept of modules

UNIT-IV

- Q8 (a) Elaborate the importance of pip command to install any library in python? (4.5)
- (b) By using NumPy library in program, perform the following operations:-
Creation of 1D array, Re-shaping of an Array, Array Slicing, Aggregate Operations. (8)
- (a) Explain file modes of file handling? Give example of each? (4.5)
- (b) Explain the use of NumPy, Pandas, Matplotlib in python programming? Give example of each? (8)

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